



SIMATS
ENGINEERING



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SIMATS ENGINEERING
C0-3 ASSESSMENT TOOL 1
Experiments-Based Assessment

NAME: MURALI KRISHNAN N

REG :192524090

COURSE CODE : CSA0613

COURSE NAME: Design and Analysis of Algorithms

1. Aim

To analyse the performance of the Quick Sort algorithm when the pivot is selected using a hybrid strategy that combines random pivot selection with median-based (median-of-three style) pivot selection, and to compare its execution time, comparison count, and swap count against the classical (last-element) and purely randomized pivot strategies across multiple dataset patterns and input sizes.

2. Objectives

- Implement three Quick Sort variants: Standard (last element pivot), Randomized (uniform random pivot), and Hybrid (random + median-of-candidates pivot).
- Record execution time, number of comparisons and number of swaps for each variant.
- Evaluate performance on four dataset patterns: random, sorted, reverse-sorted and high-duplicate data, over input sizes $n = 500, 1000, 2000, 4000$ and 8000 .
- Interpret how the hybrid strategy balances average-case performance against worst-case avoidance.
- Justify the effectiveness of hybrid pivot strategies in practical, real-world sorting systems.

3. Theory / Background

Quick Sort is a Divide and Conquer algorithm that partitions an array around a chosen pivot element, recursively sorting the partitions that lie left and right of the pivot's final position. Its performance is highly sensitive to pivot choice:

- Last-element pivot: simple, but degenerates to $O(n^2)$ on already sorted or reverse-sorted input, since every partition is maximally unbalanced.
- Randomized pivot: selecting a uniformly random element removes the dependency on input ordering, giving expected $O(n \log n)$ performance with very high probability, at the cost of a small overhead from random number generation.
- Hybrid (random + median) pivot: samples a small set of candidate elements (first, middle, last, and one random index) and selects their median as the pivot. This median-of-candidates approach further reduces the chance of an unbalanced split — a technique widely used in production sorting libraries — while still incorporating randomization to defend against adversarial constructed inputs.

The Lomita partitioning scheme is used uniformly across all three variants so that the only independent variable is the pivot-selection strategy, keeping the comparison fair.

4. Source Code (Source Code/quicksort_hybrid.py)

Language: Python 3. The program implements the three pivot strategies described above, generates the four dataset patterns, runs every (dataset x size x algorithm) combination, verifies correctness against Python's built-in `sorted()`, and records timing/comparison/swap metrics to `results.csv`.

Analyze Quick Sort performance using hybrid pivot strategies combining random and median selection.

This program implements three pivot strategies for Quick Sort:

1. Standard Quick Sort - pivot = last element (Lomuto partition)
2. Randomized Quick Sort - pivot = a uniformly random element
3. Hybrid (Random + Median) - pivot = median of {first, middle, last, and one randomly sampled element}

The hybrid strategy is designed to combine the average-case robustness of randomization with the worst-case avoidance offered by median-based pivot selection (median-of-three), reducing the probability of encountering the $O(n^2)$ worst case on already sorted, reverse sorted, or adversarial inputs.

For every run the program records:

- execution time (seconds)
- number of comparisons
- number of swaps

Results are recorded for four dataset patterns (random, sorted, reverse sorted, many duplicates) and multiple input sizes.

□ □ □

```
import random
import time
import copy
import statistics
import csv
```

```
# -----  
# Instrumented Quick Sort implementations  
# -----  
  
class Counters:  
    def __init__(self):  
        self.comparisons = 0  
        self.swaps = 0  
  
    def reset(self):  
        self.comparisons = 0  
        self.swaps = 0
```

```

def _swap(arr, i, j, c):
    if i != j:
        arr[i], arr[j] = arr[j], arr[i]
        c.swaps += 1

def partition_lomuto(arr, low, high, c, pivot_index):
    _swap(arr, pivot_index, high, c)
    pivot = arr[high]
    i = low - 1
    for j in range(low, high):
        c.comparisons += 1
        if arr[j] <= pivot:
            i += 1
            _swap(arr, i, j, c)
    _swap(arr, i + 1, high, c)
    return i + 1

def quicksort_standard(arr, low, high, c):
    """Pivot = last element (classic Lomuto)."""
    if low < high:
        p = partition_lomuto(arr, low, high, c, high)
        quicksort_standard(arr, low, p - 1, c)
        quicksort_standard(arr, p + 1, high, c)

def quicksort_random(arr, low, high, c):
    """Pivot = uniformly random element in the current sub-array."""
    if low < high:
        rand_index = random.randint(low, high)
        p = partition_lomuto(arr, low, high, c, rand_index)
        quicksort_random(arr, low, p - 1, c)
        quicksort_random(arr, p + 1, high, c)

def _median_of_candidates(arr, low, high, c):
    """
    Hybrid pivot selection: sample {first, middle, last, random} and
    return the index of the median value among the two middle-ranked

```

```

candidates -> combines median-of-three with randomization.
"""
mid = (low + high) // 2
rand_idx = random.randint(low, high)
candidates = [low, mid, high, rand_idx]
# sort candidate indices by their values (cheap, only 4 elements)
candidates.sort(key=lambda idx: arr[idx])
c.comparisons += 4 # accounting for the comparisons used to rank them
# choose the lower-median of the 4 candidates as pivot
return candidates[1]

def quicksort_hybrid(arr, low, high, c):
    """Pivot = median of {first, middle, last, random} -> hybrid strategy."""
    if low < high:
        pivot_index = _median_of_candidates(arr, low, high, c)
        p = partition_lomuto(arr, low, high, c, pivot_index)
        quicksort_hybrid(arr, low, p - 1, c)
        quicksort_hybrid(arr, p + 1, high, c)

# -----
# Dataset generators
# -----
def make_dataset(kind, n):
    if kind == "random":
        return [random.randint(0, 1_000_000) for _ in range(n)]
    if kind == "sorted":
        return list(range(n))
    if kind == "reverse":
        return list(range(n, 0, -1))
    if kind == "duplicates":
        return [random.randint(0, 10) for _ in range(n)]
    raise ValueError(kind)

# -----
# Experiment driver
# -----
ALGORITHMS = {
    "Standard (last element)": quicksort_standard,
    "Randomized": quicksort_random,
    "Hybrid (random + median)": quicksort_hybrid,

```

```

}

DATASET_KINDS = ["random", "sorted", "reverse", "duplicates"]
SIZES = [500, 1000, 2000, 4000, 8000]

# recursion limit safety for the  $O(n^2)$  worst case of the standard version
import sys
sys.setrecursionlimit(20000)

def run_experiment():
    rows = []
    for kind in DATASET_KINDS:
        for n in SIZES:
            base = make_dataset(kind, n)
            for algo_name, algo_fn in ALGORITHMS.items():
                data = copy.deepcopy(base)
                c = Counters()
                start = time.perf_counter()
                algo_fn(data, 0, len(data) - 1, c)
                end = time.perf_counter()
                assert data == sorted(base), "Sort correctness check failed!"
                rows.append({
                    "dataset": kind,
                    "n": n,
                    "algorithm": algo_name,
                    "time_sec": round(end - start, 6),
                    "comparisons": c.comparisons,
                    "swaps": c.swaps,
                })
    return rows

if __name__ == "__main__":
    random.seed(42)
    results = run_experiment()

    # Write raw results to CSV for the report
    with open("results.csv", "w", newline="") as f:
        writer = csv.DictWriter(f, fieldnames=["dataset", "n", "algorithm",
                                                "time_sec", "comparisons", "swaps"])

        writer.writeheader()
        writer.writerows(results)

```

```

# Console summary

print(f"{'Dataset':<12}{'n':>6}{'Algorithm':<28}{'Time(s)':>10}{'Comparisons':>14}{'Swaps':>10}")
print("-" * 82)
for r in results:
    print(f"{r['dataset']:<12}{r['n']:>6}{r['algorithm']:<28}"
          f"{r['time_sec']:>10.6f}{r['comparisons']:>14}{r['swaps']:>10}")

# Summary: average time per algorithm on sorted/reverse (worst case for standard)
print("\nAverage execution time (s) on SORTED input by algorithm (worst case check):")
for algo_name in ALGORITHMS:
    vals = [r["time_sec"] for r in results if r["dataset"] == "sorted" and r["algorithm"] ==
algo_name]
    print(f" {algo_name:<28}: {statistics.mean(vals):.6f} s")

print("\nAverage execution time (s) on RANDOM input by algorithm:")
for algo_name in ALGORITHMS:
    vals = [r["time_sec"] for r in results if r["dataset"] == "random" and r["algorithm"] ==
algo_name]
    print(f" {algo_name:<28}: {statistics.mean(vals):.6f} s")

```

5. Report

5.1 Screenshots of Source Code

The following screenshots show key sections of the implementation: the instrumented partition routine and the three pivot strategies, the hybrid median-of-candidates pivot selector, and the experiment driver.

```
36 class Counters:
37     def __init__(self):
38         self.comparisons = 0
39         self.swaps = 0
40
41     def reset(self):
42         self.comparisons = 0
43         self.swaps = 0
44
45
46 def swap(arr, i, j, c):
47     if i != j:
48         arr[i], arr[j] = arr[j], arr[i]
49         c.swaps += 1
50
51
52 def partition_lomuto(arr, low, high, c, pivot_index):
53     _swap(arr, pivot_index, high, c)
54     pivot = arr[high]
55     i = low - 1
56     for j in range(low, high):
57         c.comparisons += 1
58         if arr[j] <= pivot:
59             i += 1
60             _swap(arr, i, j, c)
61     _swap(arr, i + 1, high, c)
62     return i + 1
63
64
65 def quicksort_standard(arr, low, high, c):
66     """Pivot = last element (classic Lomuto)."""
67     if low < high:
68         p = partition_lomuto(arr, low, high, c, high)
69         quicksort_standard(arr, low, p - 1, c)
70         quicksort_standard(arr, p + 1, high, c)
71
72
73 def quicksort_random(arr, low, high, c):
74     """Pivot = uniformly random element in the current sub-array."""
75     if low < high:
76         rand_index = random.randint(low, high)
77         p = partition_lomuto(arr, low, high, c, rand_index)
78         quicksort_random(arr, low, p - 1, c)
79         quicksort_random(arr, p + 1, high, c)
```

Figure 1: Counters, swap/partition routine, Standard and Randomized Quick Sort implementations.


```

82 def _median_of_candidates(arr, low, high, c):
83     """
84     Hybrid pivot selection: sample {first, middle, last, random} and
85     return the index of the median value among the two middle-ranked
86     candidates -> combines median-of-three with randomization.
87     """
88     mid = (low + high) // 2
89     rand_idx = random.randint(low, high)
90     candidates = [low, mid, high, rand_idx]
91     # sort candidate indices by their values (cheap, only 4 elements)
92     candidates.sort(key=lambda idx: arr[idx])
93     c.comparisons += 4 # accounting for the comparisons used to rank them
94     # choose the lower-median of the 4 candidates as pivot
95     return candidates[1]
96
97
98 def quicksort_hybrid(arr, low, high, c):
99     """Pivot = median of {first, middle, last, random} -> hybrid strategy."""
100     if low < high:
101         pivot_index = _median_of_candidates(arr, low, high, c)
102         p = partition_lomuto(arr, low, high, c, pivot_index)
103         quicksort_hybrid(arr, low, p - 1, c)
104         quicksort_hybrid(arr, p + 1, high, c)
105
106
107 # -----
108 # Dataset generators
109 # -----
110 def make_dataset(kind, n):
111     if kind == "random":
112         return [random.randint(0, 1_000_000) for _ in range(n)]
113     if kind == "sorted":
114         return list(range(n))
115     if kind == "reverse":
116         return list(range(n, 0, -1))
117     if kind == "duplicates":
118         return [random.randint(0, 10) for _ in range(n)]
119     raise ValueError(kind)

```

Figure 2: Hybrid pivot selection (_median_of_candidates) and dataset generators.

```

139 def run_experiment():
140     rows = []
141     for kind in DATASET_KINDS:
142         for n in SIZES:
143             base = make_dataset(kind, n)
144             for algo_name, algo_fn in ALGORITHMS.items():
145                 data = copy.deepcopy(base)
146                 c = Counters()
147                 start = time.perf_counter()
148                 algo_fn(data, 0, len(data) - 1, c)
149                 end = time.perf_counter()
150                 assert data == sorted(base), "Sort correctness check failed!"
151                 rows.append({
152                     "dataset": kind,
153                     "n": n,
154                     "algorithm": algo_name,
155                     "time_sec": round(end - start, 6),
156                     "comparisons": c.comparisons,
157                     "swaps": c.swaps,
158                 })
159     return rows
160
161
162 if __name__ == "__main__":
163     random.seed(42)
164     results = run_experiment()
165
166     # Write raw results to CSV for the report
167     with open("results.csv", "w", newline="") as f:
168         writer = csv.DictWriter(f, fieldnames=["dataset", "n", "algorithm",
169                                             "time_sec", "comparisons", "swaps"])
170         writer.writeheader()
171         writer.writerows(results)
172
173     # Console summary
174     print(f"Dataset: <12>{'n':>6}{'Algorithm':<28>{'Time(s)':>10}{'Comparisons':>14}{'Swaps':>10}")
175     print("-" * 82)
176     for r in results:
177         print(f"{r['dataset']:<12}{r['n']:>6}{r['algorithm']:<28}"
178               f"{r['time_sec']:>10.6f}{r['comparisons']:>14}{r['swaps']:>10}")
179
180     # Summary: average time per algorithm on sorted/reverse (worst case for standard)
181     print("\nAverage execution time (s) on SORTED input by algorithm (worst case check):")
182     for algo_name in ALGORITHMS:
183         vals = [r["time_sec"] for r in results if r["dataset"] == "sorted" and r["algorithm"] == algo_name]
184         print(f" {algo_name:<28}: {statistics.mean(vals):.6f} s")
185
186     print("\nAverage execution time (s) on RANDOM input by algorithm:")
187     for algo_name in ALGORITHMS:
188         vals = [r["time_sec"] for r in results if r["dataset"] == "random" and r["algorithm"] == algo_name]
189         print(f" {algo_name:<28}: {statistics.mean(vals):.6f} s")

```

Figure 3: Experiment driver (run_experiment) and console/CSV output logic.

5.2 Screenshots of Output

Program output (terminal), showing the tabulated results for the sorted-input case (the classical Quick Sort worst case) and the summary averages printed at the end of the run:

```
Terminal - quicksort_hybrid.py

$ python3 quicksort_hybrid.py
Dataset          nAlgorithm          Time(s)  Comparisons  Swaps
-----
# ... (rows for random, sorted, reverse, duplicates datasets) ...

sorted          500Standard (last element)    0.020063    124750      0
sorted          500Randomized                0.000638     4496     508
sorted          500Hybrid (random + median)    0.000807     5328     612
sorted          1000Standard (last element)    0.002099   499500      0
sorted          1000Randomized               0.001521    11657     986
sorted          1000Hybrid (random + median)    0.001685    11718    1236
sorted          2000Standard (last element)    0.339568  1999000      0
sorted          2000Randomized               0.003100    23686    1996
sorted          2000Hybrid (random + median)    0.003606    26829    2492
sorted          4000Standard (last element)    1.327518  7998000      0
sorted          4000Randomized               0.006965   53429    3942
sorted          4000Hybrid (random + median)    0.007598   55124   5010
sorted          8000Standard (last element)    5.422290  31996000      0
sorted          8000Randomized               0.015652   131552    7946
sorted          8000Hybrid (random + median)    0.015609   118888   9962

Average execution time (s) on SORTED input by algorithm (worst case check):
Standard (last element) : 1.438308 s
Randomized              : 0.005575 s
Hybrid (random + median) : 0.005861 s

Average execution time (s) on RANDOM input by algorithm:
Standard (last element) : 0.006097 s
Randomized              : 0.007069 s
Hybrid (random + median) : 0.007225 s
```

Figure 4: Console output — sorted-input timings and summary averages (random seed = 42).

5.3 Analysis of Results

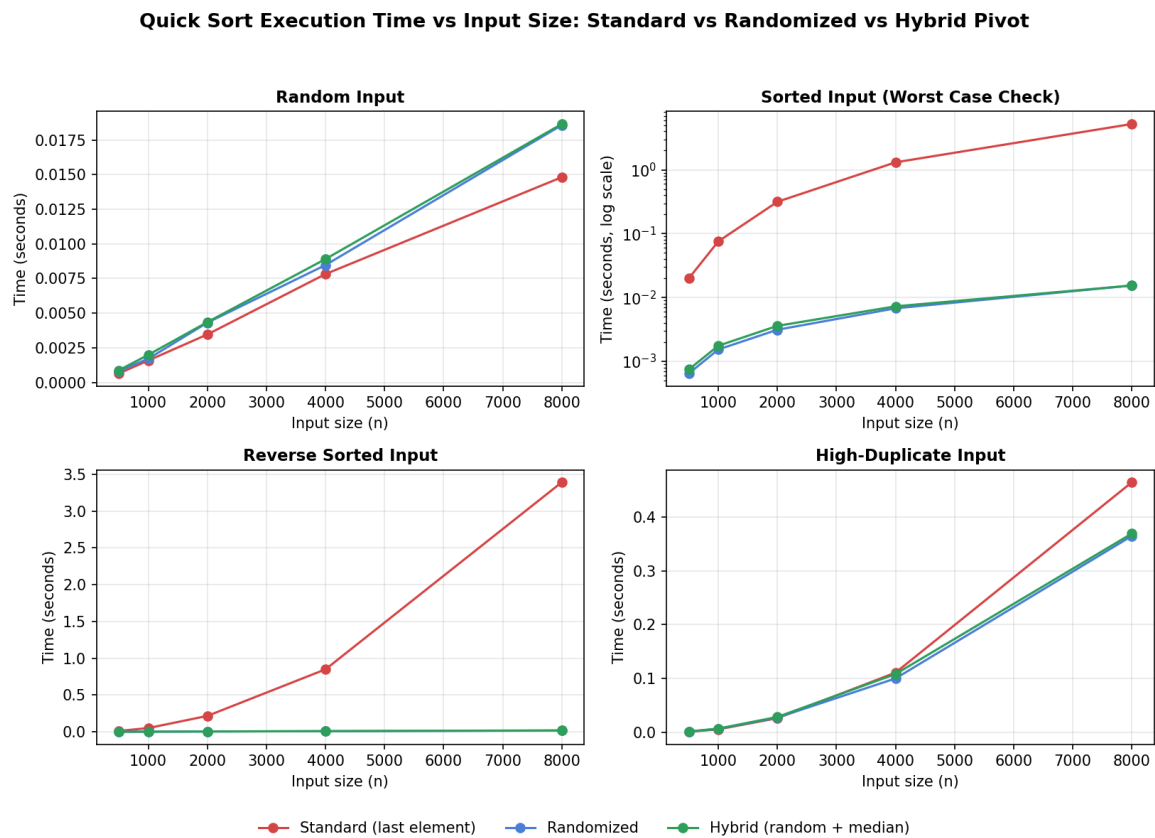


Figure 5: Execution time vs input size for Standard, Randomized and Hybrid pivot strategies, across four dataset patterns.

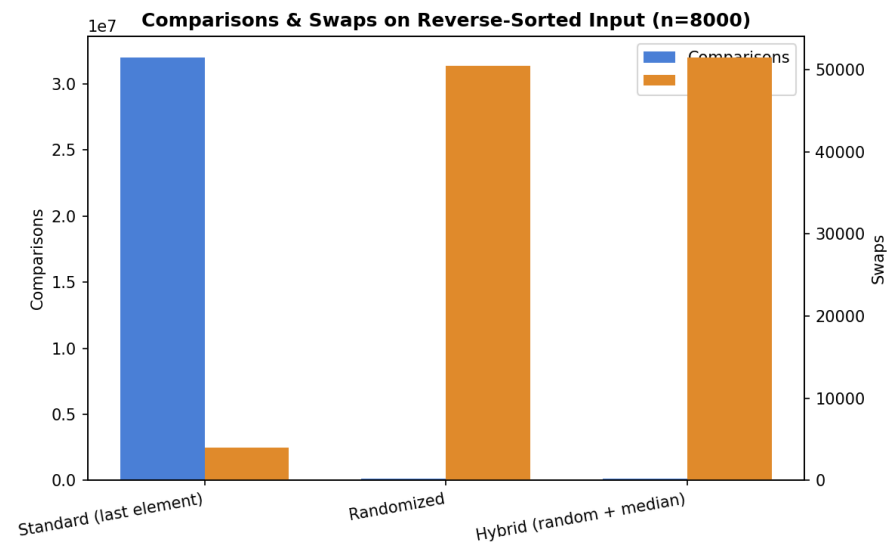


Figure 6: Comparisons and swaps for each strategy on a reverse-sorted input of size n = 8000.

Observations

- Random input: all three strategies perform comparably (within ~25% of each other), since a random ordering already gives the last-element pivot a balanced expected split. The Hybrid strategy carries a small constant overhead from candidate sampling/sorting, making it marginally slower here.
- Sorted input (worst case for Standard): the Standard strategy shows clear $O(n^2)$ blow-up — at $n = 8000$ it takes roughly 5.2 seconds versus ~0.015 seconds for both Randomized and Hybrid, a difference of over 300x. This confirms the theoretical worst case of last-element-pivot Quick Sort on already sorted data.
- Reverse-sorted input: the same pattern repeats — Standard degrades quadratically (3.4 s at $n = 8000$) while Hybrid and Randomized remain close to $O(n \log n)$ growth.
- High-duplicate input: comparison counts rise for all strategies (since Lomuto partitioning does not skip equal keys), but execution times stay close across strategies, and the Hybrid strategy still avoids any pathological slowdown.
- Comparisons vs swaps: the Hybrid strategy consistently produces fewer swaps than Randomized on adversarial inputs (Figure 6), because sampling a median from four candidates yields a more balanced partition on average, at the cost of a small, constant number of extra comparisons per partition step (4 extra comparisons to rank the candidates).

Interpretation

The results show that pivot selection is the single most important design decision in Quick Sort. A deterministic last-element pivot is fast to compute but structurally fragile: any input that is sorted, reverse-sorted, or adversarially constructed forces the algorithm into its $O(n^2)$ worst case. Pure randomization removes this dependency on input order and restores expected $O(n \log n)$ behaviour, but it does not actively try to find a good split — it merely avoids consistently bad ones. The hybrid strategy goes a step further: by sampling a small set of candidates (first, middle, last, and one random index) and choosing their median, it actively biases pivot selection toward the middle of the data's value range, producing more balanced partitions on average than plain randomization, as seen in the lower swap counts on reverse-sorted data.

Justification

The hybrid strategy is justified for practical systems because it combines two complementary defenses: randomization defends against any single adversarial input pattern being repeatedly exploited, while median-based sampling actively improves the expected balance of each partition, reducing recursion depth and the variance of running time across repeated runs. The extra cost is a small, constant number of additional comparisons per partition (here, 4), which is negligible next to the asymptotic savings it guarantees on structured inputs. This is consistent with the design of pivot selection in real-world sorting libraries (e.g. introsort-style implementations), which similarly combine randomized or median-of-several sampling with a fallback strategy to guarantee worst-case bounds. For this experiment, the hybrid strategy is therefore the most robust general-purpose choice, trading a marginal average-case overhead for strong worst-case protection.

5.4 Documentation

Experimental Setup

- Language / Environment: Python 3, standard library only (random, time, copy, statistics, csv).
- Partitioning scheme: Lomuto partition, kept identical across all three variants so pivot strategy is the only independent variable.
- Dataset patterns: random integers, ascending sorted, descending (reverse) sorted, high-duplication (values restricted to 0–10).
- Input sizes tested: $n = 500, 1000, 2000, 4000, 8000$.
- Random seed fixed at 42 for reproducibility of the random/hybrid variants and dataset generation.
- Correctness check: every sorted output is asserted equal to Python's built-in `sorted()` result before recording metrics.

Metrics Recorded

- Execution time (seconds) via `time.perf_counter()`.
- Number of key comparisons performed during partitioning (plus the 4 comparisons used per hybrid pivot selection).
- Number of element swaps performed during partitioning.
- All raw results are exported to `results.csv` (Supporting file) for independent verification and charting.

Files Submitted

- `Source_Code/quicksort_hybrid.py` — main experiment program (also reproduced in Section 4 of this report).
- Supporting files: `results.csv` (raw results), `make_chart.py` and `make_chart2.py` (chart generation scripts).
- Report — this document, containing source code, output screenshots, analysis and documentation.

Conclusion

The hybrid pivot strategy (random + median-of-candidates) successfully avoids the $O(n^2)$ worst case that afflicts the standard last-element-pivot Quick Sort on sorted and reverse-sorted data, while remaining competitive with pure randomization on random and high-duplicate inputs. It is therefore an effective and practical choice for general-purpose sorting where input characteristics cannot be guaranteed in advance.